INSTITUTO TECNOLOGICO DE LAS AMERICAS

NOMBRE ESTUDIANTE:

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MATRICULA:

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NOMBRE MAESTRO/A:

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MATERIA:

BASE DE DATOS AVANZADA

1. Tablas que empiezan con JO (HR)

SQL Worksheet

```sql
SELECT table_name
FROM all_tables
WHERE owner = 'HR'
  AND table_name LIKE 'JO%'
ORDER BY table_name;
```

TABLE_NAME
JOBS
JOB_HISTORY

Download CSV

2. Inicial + apellido de cada empleado

SQL Worksheet

```
2.
SELECT SUBSTR(first_name,1,1) || ' ' || last_name AS nombre
FROM hr.employees;
```

NOMBRE
E Abel
S Ande
M Atkinson
D Austin
H Baer
S Baida
A Banda

3. Nombre completo + email donde email contiene 'IN'

SQL Worksheet

```sql
3.
SELECT first_name || ' ' || last_name || ' ' || email AS info
FROM hr.employees
WHERE email LIKE '%IN%';
```

INFO
Mozhe Atkinson MATKINSO
David Austin DAUSTIN
Julia Dellinger JDELLING
Shelley Higgins SHIGGINS
Payam Kaufling PKAUFLIN
Janette King JKING

4. Apellido más pequeño y más grande

SQL Worksheet

```sql
4.
SELECT MIN(last_name) AS menor_apellido,
       MAX(last_name) AS mayor_apellido
FROM hr.employees;
```

MENOR_APELLIDO	MAYOR_APELLIDO
Abel	Zlotkey

5. Salario semanal entre 700 y 3000 formateado

SQL Worksheet

```
5.
SELECT TO_CHAR(salary / 4, '$9999.99') AS weekly_salary
FROM hr.employees
WHERE salary/4 BETWEEN 700 AND 3000;
```

WEEKLY_SALARY
$2250.00
$1500.00
$1200.00
$1200.00

6. Empleados con su job_title (ordenado por job_title)

SQL Worksheet

```sql
6.
SELECT e.first_name, e.last_name, j.job_title
FROM hr.employees e
JOIN hr.jobs j ON e.job_id = j.job_id
ORDER BY j.job_title;
```

FIRST_NAME	LAST_NAME	JOB_TITLE
Daniel	Faviet	Accountant
Jose Manuel	Urman	Accountant
Ismael	Sciarra	Accountant
John	Chen	Accountant

7. Job title + rango salarial del job + salario del empleado

SQL Worksheet

```
7.
SELECT j.job_title,
       j.min_salary || ' - ' || j.max_salary AS rango,
       e.salary
FROM hr.employees e
JOIN hr.jobs j ON e.job_id = j.job_id;
```

JOB_TITLE	RANGO	SALARY
Public Accountant	4200 - 9000	8300
Accounting Manager	8200 - 16000	12008
Administration Assistant	3000 - 6000	4400
President	20080 - 40000	24000
Administration Vice President	15000 - 30000	17000
Administration Vice President	15000 - 30000	17000

8. Inicial + apellido + departamento (ANSI JOIN usando todas las FK)

SQL Worksheet

```sql
8.
SELECT SUBSTR(e.first_name,1,1) AS inicial,
       e.last_name,
       d.department_name
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id;
```

INICIAL	LAST_NAME	DEPARTMENT_NAME
J	Whalen	Administration
P	Fay	Marketing
M	Hartstein	Marketing
S	Tobias	Purchasing
K	Colmenares	Purchasing

9. Igual que la anterior pero solo uniendo por department_id

SQL Worksheet

```
9.
SELECT SUBSTR(e.first_name,1,1) AS inicial,
       e.last_name,
       d.department_name
FROM hr.employees e
JOIN hr.departments d USING (department_id);
```

J	Fleaur	Shipping
A	Fripp	Shipping
T	Gates	Shipping
K	Gee	Shipping
G	Geoni	Shipping
D	Grant	Shipping

10. Apellido + (nobody/somebody) según si tiene jefe (DECODE)

SQL Worksheet

```
       d.department_name
FROM hr.employees e
JOIN hr.departments d USING (department_id);

10.
SELECT last_name,
       DECODE(manager_id, NULL, 'nobody', 'somebody') AS jefe
FROM hr.employees;
```

LAST_NAME	JEFE
King	nobody
Kochhar	somebody
De Haan	somebody

11. Inicial + apellido + salario + indicador SI/NO de si tiene comisión

```
11.
SELECT SUBSTR(first_name,1,1) || ' ' || last_name AS employee_name,
       salary AS salary,
       DECODE(commission_pct, NULL, 'No', 'Yes') AS commission
FROM hr.employees;
```

EMPLOYEE_NAME	SALARY	COMMISSION
S King	24000	No
N Kochhar	17000	No
L De Haan	17000	No
A Hunold	9000	No

12. Empleados + departamentos + ciudades (LEFT JOIN)

```sql
12.
SELECT e.last_name,
       d.department_name,
       l.city,
       l.state_province
FROM hr.departments d
LEFT JOIN hr.employees e ON e.department_id = d.department_id
LEFT JOIN hr.locations l ON l.location_id = d.location_id;
```

LAST_NAME	DEPARTMENT_NAME	CITY	STATE_PROVINCE
Hunold	IT	Southlake	Texas
Ernst	IT	Southlake	Texas
Austin	IT	Southlake	Texas
Pataballa	IT	Southlake	Texas

13. Nombre + apellido + primer valor existente: commission → manager → -1

```sql
13.
SELECT first_name,
       last_name,
       COALESCE(commission_pct, manager_id, -1) AS resultado
FROM hr.employees;
```

FIRST_NAME	LAST_NAME	RESULTADO
Steven	King	-1
Neena	Kochhar	100
Lex	De Haan	100
Alexander	Hunold	102
Bruce	Ernst	103

14. Empleados con department_id > 50 + job grade

```
14.
SELECT e.last_name,
       e.salary,
       CASE
           WHEN e.salary BETWEEN 2000 AND 5000 THEN 'A'
           WHEN e.salary BETWEEN 5001 AND 10000 THEN 'B'
           WHEN e.salary BETWEEN 10001 AND 15000 THEN 'C'
           WHEN e.salary > 15000 THEN 'D'
           ELSE 'N/A'
       END AS job_grade
FROM hr.employees e
WHERE e.department_id > 50;
```

LAST_NAME	SALARY	JOB_GRADE
King	24000	D
Kochhar	17000	D
De Haan	17000	D
Hunold	9000	B

15. Empleados sin dept. + departamentos sin empleados (FULL OUTER JOIN)

```sql

15.
SELECT e.last_name,
       d.department_name
FROM hr.employees e
FULL OUTER JOIN hr.departments d
     ON e.department_id = d.department_id;
```

LAST_NAME	DEPARTMENT_NAME
King	Executive
Kochhar	Executive
De Haan	Executive
Hunold	IT

16. Árbol jerárquico empezando por employee_id = 100

```sql
16.
SELECT last_name,
       CONNECT_BY_ROOT last_name AS jefe,
       LEVEL AS posicion
FROM hr.employees
START WITH employee_id = 100
CONNECT BY PRIOR employee_id = manager_id;
```

LAST_NAME	JEFE	POSICION
King	King	1
Kochhar	King	2
Greenberg	King	3
Faviet	King	4

17. Fecha más antigua, más reciente y número de empleados

```
17.
SELECT MIN(hire_date) AS lowest,
       MAX(hire_date) AS highest,
       COUNT(*) AS no_of_employees
FROM hr.employees;
```

LOWEST	HIGHEST	NO_OF_EMPLOYEES
13-JAN-01	21-APR-08	107

18. Departamentos con costo salarial entre 15000 y 31000

SQL Worksheet

```
18.
SELECT d.department_name,
       SUM(e.salary) AS costo
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id
GROUP BY d.department_name
HAVING SUM(e.salary) BETWEEN 15000 AND 31000
ORDER BY costo;
```

DEPARTMENT_NAME	COSTO
Marketing	19000
Accounting	20308
Purchasing	24900
IT	28800

19. Nombre del departamento, jefe y salario promedio

SQL Worksheet

```
19.
SELECT d.department_name,
       d.manager_id,
       m.last_name AS manager_name,
       AVG(e.salary) AS avg_salary
FROM hr.departments d
LEFT JOIN hr.employees m ON d.manager_id = m.employee_id
LEFT JOIN hr.employees e ON e.department_id = d.department_id
GROUP BY d.department_name, d.manager_id, m.last_name;
```

DEPARTMENT_NAME	MANAGER_ID	MANAGER_NAME	AVG_SALARY
Administration	200	Whalen	4400
Treasury	-	-	-
Public Relations	204	Baer	10000
Shareholder Services	-	-	-

20. Mayor salario promedio entre departamentos

```
GROUP BY d.department_name, d.manager_id, m.last_name;


20.
SELECT ROUND(MAX(avg_salary)) AS mayor_promedio
FROM (SELECT department_id, AVG(salary) AS avg_salary
      FROM hr.employees
      GROUP BY department_id);

```

MAYOR_PROMEDIO
19333

21. Costo mensual (sumatoria de salarios) por departamento

```
21.
SELECT d.department_name,
       SUM(e.salary) AS monthly_cost
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id
GROUP BY d.department_name;
```

DEPARTMENT_NAME	MONTHLY_COST
Sales	305100
Marketing	19000
Administration	4400
Purchasing	24900

22. Costo por job_id dentro del dept, por dept y total general (GROUPING SETS)

```
22.
SELECT d.department_name,
       e.job_id,
       SUM(e.salary) AS costo
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id
GROUP BY GROUPING SETS (
    (d.department_name, e.job_id),
    (d.department_name),
    (e.job_id),
    ()
);
```

DEPARTMENT_NAME	JOB_ID	COSTO
IT	IT_PROG	28800
Sales	SA_MAN	61000
Sales	SA_REP	244100
Finance	FI_MGR	12008

23. Igual que la anterior + mostrar si dept/job fue agrupado (GROUPING)

SQL Worksheet

```
);

23.
SELECT d.department_name,
       e.job_id,
       SUM(e.salary) AS costo,
       GROUPING(d.department_name) AS g_dept,
       GROUPING(e.job_id) AS g_job
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id
GROUP BY CUBE (d.department_name, e.job_id);
```

DEPARTMENT_NAME	JOB_ID	COSTO	G_DEPT	G_JOB
-	-	685016	1	1
-	AD_VP	34000	1	0
-	AC_MGR	12008	1	0

24. Costo por cargo dentro del departamento + costo por ciudad (GROUPING SETS)

```sql
24.
SELECT d.department_name,
       e.job_id,
       l.city,
       SUM(e.salary) AS costo
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id
JOIN hr.locations l ON d.location_id = l.location_id
GROUP BY GROUPING SETS (
    (d.department_name, e.job_id),
    (l.city)
);
```

DEPARTMENT_NAME	JOB_ID	CITY	COSTO
-	-	Toronto	19000
-	-	London	6500
-	-	Munich	10000

25. Empleados, departamentos y ciudades — NO unidos (UNION)

```sql
25.
SELECT first_name || ' ' || last_name AS nombre, department_id, NULL AS info
FROM hr.employees
UNION
SELECT NULL, department_id, department_name
FROM hr.departments
UNION
SELECT NULL, NULL, city
FROM hr.locations;
```

NOMBRE	DEPARTMENT_ID	INFO
Adam Fripp	50	-
Alana Walsh	50	-
Alberto Errazuriz	80	-
Alexander Hunold	60	-

26. Empleados que ganan más de la media de su departamento

```sql
26.
SELECT SUBSTR(e.first_name,1,1) || ' ' || e.last_name AS nombre,
       e.salary,
       d.department_name
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id
WHERE e.salary > (
    SELECT AVG(salary)
    FROM hr.employees
    WHERE department_id = e.department_id
);
```

NOMBRE	SALARY	DEPARTMENT_NAME
M Weiss	8000	Shipping
A Fripp	8200	Shipping
P Kaufling	7900	Shipping